

AC9M6M04 • Year 6 Mathematics

Angle Relationships on Lines and at Points

Use straight, full-turn and vertically opposite relationships to find unknowns

We are learning to...

Students use 180° on a straight line, 360° at a point and equality of vertically opposite angles, combining relationships without relying only on visual appearance.

Represent

Reason

Calculate

Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

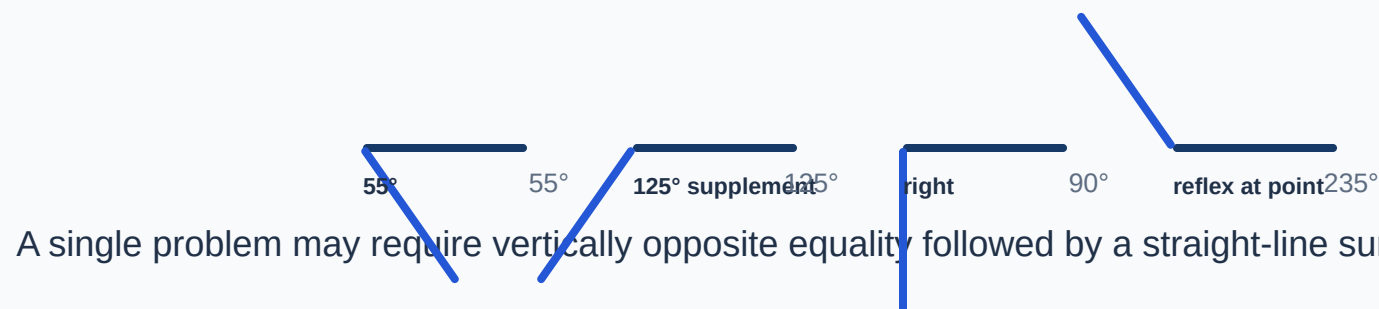
identify and use the relationships between angles on a straight line, angles at a point and vertically opposite angles to solve problems

Connect three fundamental angle relationships

relationship	equation
straight line	$a + b = 180^\circ$
angles at a point	$a + b + c + \dots = 360^\circ$
vertically opposite	$a = c$ and $b = d$
right angle	parts total 90°

The diagram may not be drawn to scale. Mark the relevant relationship and write an equation before calculating.

Solve linked unknown-angle problems



A single problem may require vertically opposite equality followed by a straight-line sum or a full-turn total.

Build and annotate the model

Represent angle relationships on lines and at points and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Diagram measured instead of relationship used:** Use given values and geometric facts.
- **Adjacent and vertically opposite confused:** Opposite angles do not share a side.
- **180° used around a point:** A full turn totals 360°.
- **Reflex angle omitted:** Include the stated turn region.

Quick check

- Find a supplement.
- Use vertical opposites.
- Sum at a point.
- Write an equation.
- Check 360° total.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.